



Compounds of Transitional elements

- Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic media. The final products obtained from it in the three conditions are, respectively
 - MnO_2, MnO_2, Mn^{2+}
 - $MnO_4^{2-}, Mn^{3+}, Mn^{2+}$
 - $MnO_2, MnO_4^{2-}, Mn^{3+}$
 - MnO, MnO_4, Mn^{2+}
- In acidic medium one mole of MnO_4^- accepts how many moles of electrons in a redox process ?
 - 1
 - 3
 - 5
 - 6
- In acidic medium potassium dichromate acts as an oxidant according to the equation, $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. What is the equivalent weight of $K_2Cr_2O_7$? (mol. Wt. = M)
 - M
 - $M/2$
 - $M/3$
 - $M/6$
- The correct formula of permanganic acid is
 - $HMnO_4$
 - $HMnO_5$
 - H_2MnO_4
 - H_2MnO_3
- Acidified potassium dichromate is treated with hydrogen sulphide. In the reaction, the oxidation number of chromium
 - Increases from + 3 to + 6
 - Decreases from +6 to +3
 - Remains unchanged
 - Decreases from +6 to +2
- When $KMnO_4$ reacts with acidified $FeSO_4$
 - Only $FeSO_4$ is oxidised
 - Only $KMnO_4$ is oxidised
 - $FeSO_4$ is oxidised $KMnO_4$ and is reduced
 - None of these
- When calomel reacts with NH_4OH , we get
 - $HgNH_2Cl$
 - $NH_2 - Hg - Hg - Cl$
 - Hg_2O
 - HgO
- $AgCl$ dissolves in a solution of NH_3 but not in water because
 - NH_3 is a better solvent than H_2O
 - Ag^+ forms a complex ion with NH_3
 - NH_3 is a stronger base than H_2O
 - The dipole moment of water is higher than NH_3
- In solid $CuSO_4 \cdot 5H_2O$ copper is co-ordinated to
 - Five water molecules
 - Four water molecules
 - One sulphate anion
 - One water molecule
- A white powder soluble in NH_4OH but insoluble in water is
 - $BaSO_4$
 - $CuSO_4$
 - $PbSO_4$
 - $AgCl$
- Verdigris** is



- (a) Basic copper acetate
(b) Basic lead acetate
(c) Basic lead
(d) None of these
12. Number of moles of $K_2Cr_2O_7$ reduced by one mole of Sn^{2+} ions is
(a) $1/3$ (b) 3
(c) 1.6 (d) 6
13. Which one of the following is reduced by hydrogen peroxide in acid medium
(a) Potassium permanganate
(b) Potassium iodide
(c) Ferrous sulphate
(d) Potassium ferrocyanide
14. Which oxide of manganese is amphoteric
(a) MnO_2 (b) Mn_2O_3
(c) Mn_2O_7 (d) MnO
15. Which one of the following oxides is ionic
(a) MnO (b) Mn_2O_7
(c) CrO_3 (d) P_2O_5
16. Correct formula of **calomel** is
(a) Hg_2Cl_2 (b) $HgCl_2$
(c) $HgCl_2 \cdot H_2O$ (d) $HgSO_4$
17. One of the important use of ferrous sulphate is in the
(a) Manufacture of blue black ink
(b) Manufacture of chalks
(c) Preparation of hydrogen sulphide
(d) Preparation of anhydrous ferric chloride
18. Copper sulphate is not used
(a) In electrotyping
(b) In dyeing and calicoprinting
(c) In detecting water
(d) As fertilizer
19. **Blue vitriol** is
(a) $CuSO_4$
(b) $CuSO_4 \cdot 5H_2O$
(c) Cu_2SO_4
(d) $CuSO_4 \cdot H_2O$
20. A solution of copper sulphate may be kept safely in the container made up of
(a) Fe (b) Ag
(c) Zn (d) Al

